

An Integrated Risk-Aware Decision Support System for Formula 1 Race Strategy Using Machine Learning and Monte Carlo Simulation

Amisha Sanjay Kadukar
M.Sc. Computing (Data Analytics)
Dublin City University
Dublin, Ireland
amishasanjay.kadukar2@dcu.mail.ie

Divyansh Doshi
M.Sc. Computing (Data Analytics)
Dublin City University
Dublin, Ireland
divyansh.doshi2@mail.dcu.mail.ie

Abstract—Race strategy in Formula 1 requires repeated decisions about pit timing, tyre choice and risk while the future state of the race remains uncertain. This paper develops an integrated decision support system that combines driver-lap predictive models with Monte Carlo simulation and a risk-aware recommendation layer. Driver-lap data from the 2019-2025 seasons were used to estimate pace, disruption risk, pit behaviour, compound choice, traffic and rejoin cost. Two results stand out: lap-time MAE was reduced by 81.64%, and rejoin or traffic-penalty MAE was reduced by 60.16% against the selected baselines. Controlled ablation showed that recent pace, circuit context, tyre state and local traffic were important inputs, while removing key simulation or risk terms changed selected strategies. The system then converted these estimates into feasible pit and wait actions, returning expected finish, expected points, ranking probabilities, downside information and confidence. Its contribution lies in this auditable integration, not in a claim of numerical superiority over studies with different data and evaluation settings.

Index Terms—Formula 1, sports analytics, machine learning, tyre strategy, Monte Carlo simulation, decision support, risk-aware optimisation.

I. INTRODUCTION

Formula 1 strategy is a sequential decision problem. At every decision lap, a team must decide whether to retain track position, change tyres, react to a rival, or wait for more information. The value of each action depends on tyre age, pace, traffic, weather, track status, pit-lane loss and the number of laps remaining. These quantities evolve together. Consequently, a decision that reduces expected lap time can still be poor if it causes the driver to rejoin in traffic or creates a large probability of losing points.

Existing research has shown that optimisation, simulation and machine learning can all improve race-strategy decisions. Dynamic programming has been used for pit-stop optimisation, Monte Carlo simulation has been used to represent race uncertainty, and reinforcement learning has been applied to sequential strategy decisions [1]–[8]. However, these methods usually address one part of the strategy problem. Prediction alone does not decide the best action, while a deterministic

optimiser cannot fully account for uncertain race interruptions, weather or rejoin conditions.

This paper develops a Formula 1 race strategy decision support system rather than a single end-to-end predictor. Its unique implementation is a shared driver-lap table that carries the outputs of Models 1-4 into Models 5A, 5B and 6. Models 1-4 estimate pace, disruption, pit and compound behaviour, and traffic or rejoin effects. Model 5A estimates local pit and wait costs, Model 5B simulates feasible actions, and Model 6 converts the simulation result into a risk-aware recommendation. The proposed system is expected to support a strategist, not replace one. It provides the expected consequence of each candidate action along with an explanation of the uncertainty that affected the recommendation.

The work makes three contributions. First, it implements a lap-level pipeline in which Models 1-4 are merged into one strategy-ready dataset rather than being used as isolated predictions. Second, it evaluates stay-out, pit-now and delayed-pit actions through local cost estimation and Monte Carlo simulation, then separates the raw expected-best action from the risk-adjusted recommendation. Third, it reports direct baseline comparisons and controlled ablations that identify the feature groups and pipeline components that materially affect predictions or final strategy selection. This paper does not claim that its metrics are directly better than every cited study, because the cited studies use different data, targets and evaluation settings.

II. LITERATURE REVIEW

Recent race-strategy research has followed three main directions: optimization under a specified race model, stochastic simulation of race uncertainty, and reinforcement learning in a simulator. Dynamic programming has been applied to Formula 1 pit timing and compound sequence [1]. Monte Carlo simulation has been used to represent uncertain lap time, pit duration and neutralisations [2], [3]. Reinforcement-learning approaches have learnt race policies from repeated simulated episodes [4]–[8].

Carrasco Heine and Thraves optimize Formula 1 pit-stop strategy through dynamic programming [1]. Heilmeier *et al.* model probabilistic race influences through Monte Carlo simulation and later use neural networks for pit and compound decisions inside a simulator [2], [9]. Paparusso *et al.* model competitor interaction through Monte Carlo simulation and stochastic dynamic programming, while Duhr *et al.* optimize lap-level hybrid power-unit energy allocation [10], [11]. These studies indicate that uncertainty and interaction matter, although their main target is either a specified race model or energy management rather than a public driver-lap pit-and-rejoin decision system.

Recent Formula 1 studies have focused on individual parts of the strategy problem rather than the complete decision process. Sasikumar *et al.* use public telemetry and deep-learning models to support pit-stop decisions [12]. Aguad and Thraves model two-driver pit-stop competition through dynamic programming and game theory [13], while Bekker and Lotz use discrete-event simulation to plan Formula One race strategies [14]. Piccinotti *et al.* examine online planning for lap-by-lap pit-stop and compound decisions [15]. Together, these works support the need to combine timing, uncertainty and action selection; they do not provide a directly comparable benchmark because their data, objectives and simulation settings differ from this project.

Formula E work provides a useful learning-based comparison. Liu and Fotouhi combine neural-network prediction with Monte Carlo tree search [4]; Liu *et al.* use distributed policy-gradient reinforcement learning and multi-car reinforcement learning [5], [6]. More recently, Thomas *et al.* study explainable reinforcement learning for Formula 1 and its generalization within a simulator [7], [8]. These methods learn an action policy in a simulator. In contrast, this project preserves separate pace, disruption, pit, compound and traffic estimates before ranking a feasible set of actions. The estimates remain inspectable by a strategist.

Direct numerical ranking against prior studies would be invalid because championships, inputs, targets and simulators differ. The comparison in Table I is therefore methodological. The proposed system combines public lap-level prediction, uncertainty-aware simulation, local traffic and rejoin cost, and risk-aware recommendation. Its ablation results then test whether these inputs materially affected the implemented predictions and decisions.

Table I maps differences in research scope and system design, not a direct accuracy ranking. The proposed pipeline addresses an integration gap by carrying outputs from pace, disruption, pit, compound and traffic models into local cost estimation, scenario simulation and a risk-aware recommendation.

III. DATASET

The analysis used historical Formula 1 data from the 2019–2025 seasons. The unit of analysis was a driver-lap, so each record described one driver’s race state before the next strategic decision. Race identifiers were retained throughout

data construction and evaluation to prevent information from the same race being shared between development and held-out data.

The dataset combines variables that change at different speeds. Race and circuit identifiers remain fixed within an event; weather and track status can change during a race, while position, gaps, tyre age and recent lap time change at the driver-lap level. Recorded fields include race and driver identifiers, lap number, current position, stint, compound, tyre age, current and previous valid lap time, laps remaining, pit-stop information, gaps ahead and behind, weather indicators, air and track temperature, rainfall and track status. The strategy-ready table also retains time-safe outputs from Models 1-4: expected lap time, fresh-tyre pace, degradation delta, disruption probabilities, pit and compound probabilities, clean-air status, position delta and rejoin penalty.

This structure is necessary because a strategy decision cannot be represented by final race position alone. It depends on the current state of the driver, tyres, nearby cars and circuit conditions. The later simulation models read the combined driver-lap state rather than future pit stops, future compound choices or final race outcomes.

Figure 1 is a data-coverage check. It shows the available driver-lap rows by season and race round, with the most populated cells containing approximately 700 to 1,400 rows. Blank cells reflect calendar differences, while the cell with 20 rows identifies an abnormal shortened-race case for review during data preparation. This coverage check makes uneven race representation visible before race-level splitting and evaluation. It does not measure strategy quality; rather, it documents the scope and irregularities of the multi-season, lap-level data used by the predictive and simulation models.

IV. METHODOLOGY

A. Problem Definition and Decision Objective

At decision lap t , the driver state includes current position, lap number, laps remaining, tyre compound and age, recent pace, gaps to nearby cars, weather, track status and circuit context. A candidate action is defined by a wait horizon and a target compound. Examples include staying out, pitting now for hard tyres, or waiting three laps before switching to intermediates. For every feasible action, the simulator produces a set of possible remaining-race outcomes by varying pace, disruption, traffic and weather conditions according to the project models.

The final recommendation was based on three recorded quantities: expected outcome, downside risk and confidence. Expected outcome was reported as expected finish, expected points or another pre-defined project outcome measure. Downside risk was reported using a declared lower-tail measure, such as low-point probability or a poor-finish percentile. Confidence reflected the reliability of the upstream probability and feasibility estimates. The implementation recorded the decision rule and any weights used to compare these quantities. This paper does not claim that a generic formula is universally

TABLE I
STATE-OF-THE-ART STRATEGY METHODS AND THE PROPOSED SYSTEM

Study	Main method	Primary focus	Difference in the proposed system
Carrasco Heine and Thraves [1]	Dynamic programming	Pit timing and compound sequence	Uses separately estimated pace, disruption, pit and rejoin inputs before action simulation.
Heilmeyer <i>et al.</i> [2], [9]	Monte Carlo simulation and neural networks	Probabilistic simulation; pit and compound decisions	Carries model outputs in one driver-lap state and tests their contribution through ablation.
Paparusso <i>et al.</i> ; Duhr <i>et al.</i> [10], [11]	Stochastic optimisation; nonlinear programming	Competitor-aware and energy strategy	Focuses on public Formula 1 driver-lap strategy, including pit and rejoin cost.
Liu <i>et al.</i> ; Thomas <i>et al.</i> [4]–[8]	MCTS and reinforcement learning	Learned policies in simulation	Retains separate predictive outputs and ranked-action explanations before the final recommendation.
Proposed system	Modular ML, simulation and ranking	Lap-level strategy support	Integrates uncertainty, pit cost, compound feasibility, traffic effects and explanation in one auditable workflow.

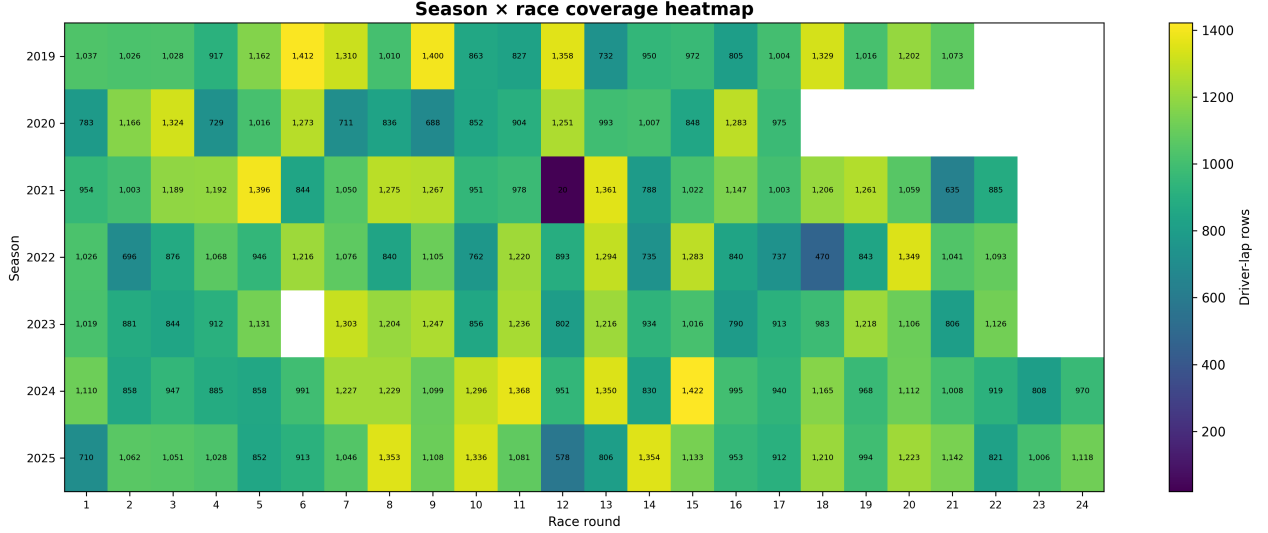


Fig. 1. Season and Race-Round Coverage of the Driver-Lap Dataset

correct; the decision rule was an implementation choice evaluated against the documented baselines and scenarios.

B. Data Preparation and Leakage Control

The unit of analysis was a driver-lap. Historical races from 2019-2025 were transformed into a master table containing race and driver identifiers, lap number, position, stint, tyre compound, tyre age, lap time, gaps, weather, track status and pit information. Additional features represented clean air, local traffic density, overtaking difficulty, estimated pit-lane loss and laps remaining.

Data quality was important because neutralised laps did not measure normal race pace. Safety Car, Virtual Safety Car and red-flag periods were marked explicitly and handled separately during pace modelling. Tyre labels and identifiers were standardized; invalid timing observations, retirements and missing values were recorded rather than silently converted into performance signals. Feature construction was bounded by the decision time. In particular, all values used at t were available before the action at t was taken. This prevented target leakage from future laps, final finishing position or post-race information.

Table II separates recorded race information from features and model outputs derived at the decision lap. This distinction matters because Models 5 and 6 use only the current strategy-ready state when generating and ranking candidate actions. Future pit stops, future tyre choices and final race outcomes were excluded from the input data, reducing the risk of future information entering a prediction. The table also shows how raw race records, contextual information and model outputs were combined into one consistent input for simulation. This structure allowed each later strategy decision to be traced back to information available at that point in the race.

Temporal validation was required. Training and test observations were not randomly mixed across future and earlier races. A season-forward split was used to tune models on earlier seasons, select them on a validation season and report final results on later held-out races. This design represented deployment, where a strategist learns from the past and applies the system to an unseen event.

C. Feature Engineering and Candidate Strategies

Feature engineering is guided by the race decision rather than by the availability of columns. Pace features include recent valid lap times, predicted clean-air pace, compound,

TABLE II
DATA USED TO CONSTRUCT THE STRATEGY-READY DRIVER-LAP TABLE

Data component	Coverage and unit	Information retained at the decision lap	Main use in the system
Historical race records	2019–2025; driver-lap	Race and driver identifiers, lap number, position, stint, tyre compound, tyre age, lap time, gaps and pit information.	Pace, degradation, pit behaviour, traffic and position modelling.
Race-context records	Matched to the driver-lap state	Circuit, weather, track status, rainfall, air and track temperature, and Safety Car, VSC or Red Flag state.	Disruption, compound-feasibility and scenario modelling.
Derived strategy state	One record per decision lap	Recent valid pace, laps remaining, local traffic density, clean-air indicator, overtaking difficulty and estimated pit-lane loss.	Candidate-action generation and local pit or wait-cost estimation.
Model-output state	Generated from Models 1–4	Pace and degradation estimates, disruption probabilities, pit and compound probabilities, and traffic or rejoin estimates.	Monte Carlo simulation and final recommendation.

tyre age and stint number. Race-progress features include current lap, laps remaining, stop count and race phase. Interaction features include current position, gaps ahead and behind, local pack density, clean-air status and estimated overtaking difficulty. Environment features include circuit, track temperature, air temperature, rainfall and track status. A useful implementation keeps a data dictionary that records the definition, time availability and imputation rule for every feature.

Strategy options are generated from the observed state rather than enumerating every theoretical tyre sequence. The options described in the project summary are stay out, pit immediately, or wait one, three or five laps, each paired with a feasible dry or wet compound. Options that violate tyre rules, exceed remaining laps, or depend on an implausible weather state are removed before simulation. This restricted set keeps the recommendation understandable to a race strategist.

For each strategy option, the system records the information that led to its inclusion or exclusion. For example, an intermediate-tyre option may be excluded in a dry base scenario or retained in a rain-risk scenario. This audit trail is needed because a recommendation is a decision between alternatives, not merely a high-probability label.

D. Model Development

Fig. 2 presents the overall decision process. Cleaned driver-lap data are first processed by Models 1–4 and then passed to Models 5A, 5B and 6 for cost estimation, simulation and final strategy recommendation. Each model was developed as a separate task with a distinct target, training dataset, validation metric and failure mode. This modular design limited the effect of an inaccurate component while enabling systematic ablation studies to evaluate the contribution of each model within the overall pipeline.

The workflow in Fig. 2 shows that Models 1–4 provide parallel predictive estimates, which are merged with the current driver-lap state before cost estimation and simulation. Each stage retained an independently testable output before it was passed to the next stage.

Model 1 was trained on valid racing laps and estimated expected lap time under normal conditions together with tyre-

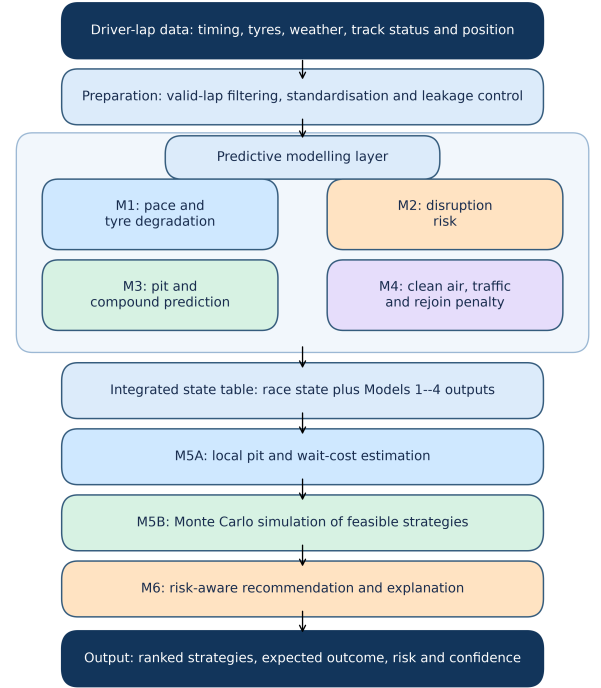


Fig. 2. Flowchart of the Strategy Decision Process

age-related pace loss. Development compared ridge regression with the selected XGBoost model [16]. Inputs included compound, tyre age, stint, circuit, recent lap time, track temperature and traffic features. The output contained expected pace, fresh-tyre counterfactual pace, degradation delta and an uncertainty estimate. MAE and RMSE were calculated on held-out races and were stratified by compound and race condition. XGBoost was selected because it handled the nonlinear relationships between tyre state, circuit conditions, traffic and pace more accurately than the linear baseline during development.

Figure 3 shows that average degradation generally increased with tyre age for the dry compounds. The sharp values at early tyre ages were retained because lap-level records include noisy

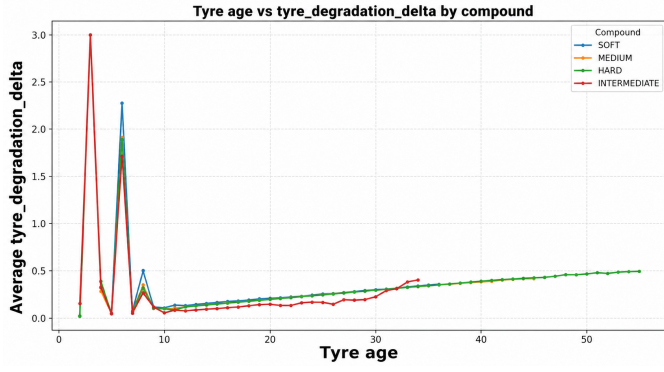


Fig. 3. Tyre Age and Average Tyre-Degradation Delta by Compound

contexts such as pit laps, neutralised laps, wet transitions and small compound-age groups. The graph is therefore evidence of a measurable tyre-age signal, not a claim of a perfectly smooth physical degradation curve. It supports the use of tyre age, compound and stint variables in Models 1 and 3.

Model 2 was developed as three hazard models for Safety Car, Virtual Safety Car and Red Flag events. Each model estimated event probability over the next three, five and ten laps from the current race state. Inputs included race phase, field spread, number of drivers still running, weather, recent event flags, circuit characteristics and gaps between cars. Logistic regression provided an interpretable probability baseline, while XGBoost captured non-linear interactions. When Red Flag data were too sparse for a stable supervised model, the system used a calibrated fallback or prior-based estimate. ROC-AUC, PR-AUC, Brier score and reliability plots were used because probabilities passed into simulation needed both calibration and discrimination.

Model 3 contained three linked components. Model 3A estimated pit probability over the next one, three and five laps. Model 3B estimated probabilities for the dry compounds, and Model 3C handled intermediate and wet-tyre logic. XGBoost classifiers were trained using tyre age, laps remaining, position, stint, current compound, weather, race phase and neutralisation probabilities. The outputs were used as feasibility and confidence signals, not as automatic commands. A strategy using a very low-probability compound could still be simulated, but received a confidence penalty. Pit probability was evaluated with ROC-AUC and PR-AUC, while compound choice used log loss and top-one and top-two accuracy.

Model 4 contained a clean-air classifier, a position-delta model and a rejoin or traffic-penalty model. XGBoost was used because traffic effects depend on interactions among position, gaps, local traffic density, pack density, overtaking difficulty, pit-lane loss, weather, expected pace and tyre state. Model 4A estimated whether the driver had clean air. Model 4B estimated short-horizon position change. Model 4C estimated time lost after a stop when the driver rejoined traffic. These outputs made the expected positional cost of a pit stop explicit in the simulator.

The development plan is summarized in Table III.

Table III summarises how the system was developed as

a set of connected but distinct analytical tasks. Models 1–4 transformed the current driver-lap state into pace, disruption, pit, compound and traffic estimates. Models 5 and 6 then used these outputs to simulate and rank feasible strategies. The table also explains why one evaluation measure was not used for every model: regression tasks required error measures, rare-event models required discrimination and calibration measures, and simulation models required decision-output and stability checks. This task-specific design made the pipeline easier to evaluate and to test through ablation.

E. Simulation and Recommendation Logic

Model 5A created local estimates of immediate pit loss and wait cost. Pit loss was circuit and context dependent, so a global mean was used only as a fallback. Model 5B then simulated staying out, pitting immediately, waiting one lap, waiting three laps or waiting five laps, with feasible tyre compounds. Each simulated trajectory combined predicted pace, tyre degradation, uncertainty, disruption probability, traffic effects, local pit cost and weather logic. The simulator applied dry-to-wet and wet-to-dry crossover penalties and tyre warm-up penalties so that inappropriate compounds were not rewarded.

The simulation produces a distribution for each candidate action: expected finish position, expected points, probability of finishing in the top ten, top five, podium or win, and a downside statistic such as the 10th percentile of points or finish position. The final model does not blindly choose the lowest mean cost. It uses the recorded expected outcome, downside measure and confidence rule, flags weak confidence, and produces a short explanation containing the main pace, traffic, weather and risk factors.

The simulator used common random numbers when comparing actions within the same decision state. Competing actions were evaluated with the same sampled disruption and pace shocks where appropriate, reducing Monte Carlo noise relative to independently sampled action scores. The number of runs was selected through a convergence check, which repeated the simulation at increasing run counts and checked the stability of the action ranking and reported uncertainty. The implementation recorded the run count, seed policy and computation time per decision state.

The primary output was a ranked strategy table, not a single command. For each candidate, the table showed wait horizon, target compound, expected finish, expected points, top-ten probability, selected downside measure, confidence and the main explanation. For example, the explanation could state that pitting immediately gained fresh-tyre pace but incurred a likely traffic rejoin, whereas staying out protected track position and had lower downside. This format enabled a human to disagree with the recommendation while still using the analysis.

The recommendation layer compared feasible actions after the simulation stage. It reported expected finish or points, a declared downside measure and confidence rather than presenting a single unexplained score. The final decision rule was

TABLE III
MODEL DEVELOPMENT PLAN: INPUTS, TRAINING TARGET AND EVALUATION MEASURES

Model	Goal and target	Development data and inputs	Primary evaluation measures
M1	Estimate valid lap time, expected pace and tyre-degradation loss.	Compound, tyre age, stint, recent valid pace, circuit, weather and traffic context.	MAE and RMSE for pace; error by compound, circuit and race condition.
M2	Estimate the probability of a Safety Car, Virtual Safety Car or red flag over a defined horizon.	Track status history, race phase, circuit, weather and incident-related context available at the decision lap.	PR-AUC, ROC-AUC, Brier score and calibration curve.
M3	Estimate pit-stop probability and the plausibility of each tyre compound.	Tyre state, laps remaining, race phase, weather, track status and strategic context.	PR-AUC or F1 for pit timing; macro-F1 and log loss for compound choice.
M4	Estimate clean-air status, rejoin penalty and position change after a stop.	Position, gaps, local traffic density, overtaking difficulty, pit-loss estimate and circuit context.	MAE and RMSE for time or position penalty; scenario-consistency checks.
M5–M6	Simulate candidate actions and rank them by expected utility, risk and confidence.	Outputs of M1–M4, action definition, weather scenario and risk appetite.	Expected points or finish, downside measure, action stability and comparison with strategy baselines.

recorded with the run output so that the reason for selecting one action over another could be checked during evaluation.

The recommendation was generated only after the data checks and model outputs had been completed. The current driver-lap state was checked for missing critical values and implausible tyre or timing values. If a critical feature was unavailable, the system reported insufficient information or used a documented fallback rather than issuing a confident recommendation. Feasible actions were then evaluated under repeated race futures.

Model 6 was a decision engine rather than another predictive model. It read the Model 5B strategy results and classified the driver as front-running (P1–P3), points-fighting (P4–P10) or a recovery drive (outside P10). It then combined expected points, expected finish, ranking probabilities, downside risk, tail-time risk and compound confidence using the selected conservative, balanced or aggressive risk appetite. Confidence depended on the utility and expected-points gap between the best and second-best strategies. This allowed Model 6 to prefer a safer action when the simulated performance difference was small but the downside or compound risk was higher.

The explanation layer distinguished evidence from speculation. It could state that a candidate’s expected penalty was caused by an estimated rejoin into traffic, or that its confidence was reduced by uncertain disruption risk. It did not present a simulated Safety Car as if it were an observed future event. This restrained wording was essential for a tool interpreted by non-specialists as well as analysts.

The implementation used a master driver-lap table containing recorded race fields, engineered features and time-safe outputs from Models 1–4. Models 5A, 5B and 6 read this shared state rather than reloading earlier models separately. The simulator accepted the race and driver identifiers, decision lap, scenario, simulation count and risk appetite. It stored both the raw simulated action ranking and the final recommendation with its confidence and explanation. This preserved the decision path for inspection without requiring a separate implementation section.

F. Experimental Setup and Evaluation Goals

The experiment evaluates whether a modular decision-support pipeline improves on simple race-strategy baselines while remaining understandable to a human strategist. The input is a driver-lap state from a held-out race. The output is a ranked set of feasible actions with a predicted outcome distribution and a risk-aware recommendation.

The data are split by race identifier, not by individual lap. This prevents laps from the same race appearing in both training and evaluation data. At a decision lap, each model uses only information available up to that lap. For example, previous lap time, tyre age, current position and current gaps are valid inputs, whereas future pit stops, future compound choices and the final race result are not. This design prevents future race information from leaking into a prediction.

The setup addresses three goals. First, it tests whether the component models predict pace, disruption, pit behaviour, compound choice and traffic effects better than transparent baselines. Second, it checks whether probability outputs are suitable for use in simulation. Third, it tests whether the full pipeline produces feasible and understandable strategy outputs under dry, wet and neutralised-race scenarios.

The methodology also supports the later comparison with the state of the art. Prior studies use optimisation, simulation or reinforcement learning under different championships, data sources and target definitions. For this reason, the paper compares system design and decision-support capability rather than incompatible accuracy values. The ablation study complements the baseline evaluation by removing feature groups or decision components and measuring the resulting loss in predictive performance or change in recommended strategy. Together, these tests show whether the individual models improved on their baselines and whether the integrated system used the information it claimed to use.

V. PERFORMANCE EVALUATION

The evaluation had two levels. Each predictive model was compared with its baseline on held-out races, while the full decision pipeline was assessed under controlled scenarios. Race-

level separation was retained throughout, and final models were frozen independently of their research baselines. Within a scenario, competing actions used the same decision state and uncertainty draws where appropriate, so differences reflected the action rather than simulation noise.

Safety Cars, Red Flags and weather transitions were uncommon, so test data retained their natural frequency and alert thresholds were selected on validation data. Scenario checks covered dry, wet and neutralised races. Each case recorded circuit, decision lap, position, tyre state, gaps, weather and track status, then reported the candidate actions and their simulated outputs. This checked that the system did not recommend clearly implausible actions, while keeping recorded race state separate from model estimates and simulated outcomes.

Each module is compared against a baseline that reflects what would be available without the final model. Model 1 is compared with ridge regression and is assessed with MAE. Models 2A–2C use historical priors and are assessed with ROC-AUC and PR-AUC. Model 3A uses a tyre-age prior and the same classification measures. Model 4B uses a no-change baseline, while Model 4C uses a historical grouped mean; both are assessed with MAE, RMSE and R^2 . The exact baseline and final values are reported in Table IV. Models 3B, 3C, 4A, 5A, 5B and 6 are reported using their measured outputs because a direct baseline comparison was not supplied for those components.

Regression was evaluated on valid racing laps using MAE and RMSE. Rare-event classification used PR-AUC alongside ROC-AUC [17]. For Models 2 and 3, Brier score and reliability curves checked whether probability estimates matched observed frequencies before they were passed into simulation.

Decision evaluation requires a different perspective. Where historical counterfactual outcomes are unavailable, the paper does not claim that an unobserved alternative would certainly have won. It instead reports agreement with known safe or unsafe scenario constraints, comparative simulated utility under a fixed scenario definition, and retrospective assessments using only information available at the decision lap. These checks provide evidence about decision consistency without presenting simulation as ground truth.

Table IV reports the available baseline comparisons. Baseline values are included only for predictive tasks for which a meaningful non-model alternative was evaluated. Alongside numerical performance, scenario checks were defined around logical strategy constraints: wet-tyre selection in a dry race required supporting evidence; neutralisation reduced pit cost without guaranteeing a stop; and an ordinary dry-race midfield state required a plausible podium probability.

In Table IV, each final predictive model is compared with the baseline selected for its task. Percentages are reported only for directly comparable error or AUC measures with a meaningful baseline scale. For rare disruption events, absolute AUC gains are reported because percentage changes from very small PR-AUC priors would be misleading. Models 5A, 5B and 6 are excluded because they produce simulated decision outputs rather than a directly comparable predictive target.

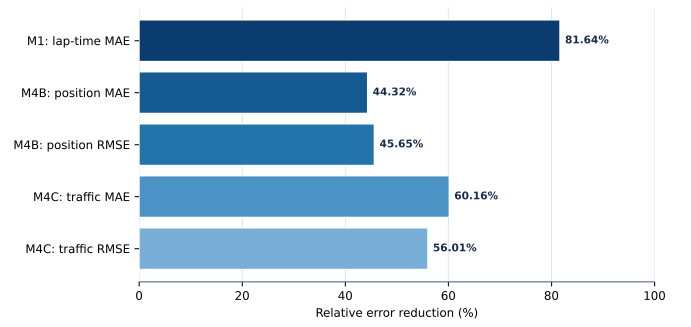


Fig. 4. Relative Error Reduction of Final Regression Models

The ablation study complemented the baseline comparisons. For supervised components, an entire feature group was permuted or removed and the held-out change in MAE, ROC-AUC, PR-AUC, log loss, accuracy or R^2 was recorded. A larger deterioration indicated a greater contribution from that group. For Model 3C, the wet-condition rule layer was checked by confirming that dry compounds did not remain the highest-probability choice on wet-handler rows.

For Models 5A and 5B, upstream model outputs or simulation components were neutralised and the resulting recommendation change was recorded. Model 6 was evaluated by removing individual utility or risk-protection terms from the same selected test case. These analyses do not estimate causal effects outside the evaluated data and scenario. They show whether the implemented system relies on the components it claims to use, and whether removing a component produces a meaningful loss of predictive information or a change in the selected strategy.

VI. RESULTS

The measured predictive comparisons are reported in Table IV. One result deserves emphasis. Lap-time MAE fell from 5.7465 for ridge regression to 1.0548 for the final XGBoost model, an 81.64% reduction. The final model followed changes in lap pace much more closely than the ridge-regression baseline.

All disruption models improved on their historical-prior baselines in both ROC-AUC and PR-AUC. For Safety Car prediction, ROC-AUC rose from 0.5319 to 0.8237 and PR-AUC from 0.0253 to 0.1303. Yet this was the harder part of the problem. Red Flags were rare, so their probabilities remained uncertain even after the model improved. Disruption estimates were used to create and weight simulated scenarios, not to claim that an interruption would certainly occur.

The pit-probability model improved on the tyre-age rule: ROC-AUC increased from 0.7380 to 0.8092 and PR-AUC from 0.2180 to 0.2895. Position-delta MAE fell from 2.4776 to 1.3795. More strikingly, rejoin-penalty MAE fell from 8.7578 to 3.4895, a 60.16% reduction. This suggests that a pit stop cannot be assessed from pit-lane time alone; where the driver rejoins, and who is nearby, can change the decision.

The bars in Fig. 4 show the reduction in prediction error relative to the corresponding baseline. Lap-time prediction

TABLE IV
PREDICTIVE MODEL COMPARISON AGAINST BASELINES

Model	Component / task	Baseline	Metric	Baseline result	Final-model result	Change from baseline
M1	Lap-time prediction	Ridge regression	MAE	5.7465	1.0548	81.64% lower MAE
M2A	Safety Car hazard	Historical prior	Average ROC-AUC	0.5319	0.8237	+0.2918
			Average PR-AUC	0.0253	0.1303	+0.1050 AUC
M2B	VSC hazard	Historical prior	Average ROC-AUC	0.5074	0.7063	+0.1989
			Average PR-AUC	0.0063	0.0954	+0.0891 AUC
M2C	Red Flag hazard	Historical prior	Average ROC-AUC	0.1509	0.8481	+0.6972
			Average PR-AUC	0.0008	0.0637	+0.0629 AUC
M3A	Pit probability	Tyre-age prior	Average ROC-AUC	0.7380	0.8092	ROC +9.65%
			PR-AUC	0.2180	0.2895	PR +32.80%
M4B	Position delta	No-change baseline	MAE	2.4776	1.3795	MAE -44.32%
			RMSE	3.5510	1.9299	RMSE -45.65%
			R^2	-0.8695	0.4478	R^2 +1.3173
M4C	Rejoin / traffic penalty	Historical grouped mean	MAE	8.7578	3.4895	MAE -60.16%
			RMSE	13.0142	5.7249	RMSE -56.01%
			R^2	-0.0156	0.7198	R^2 +0.7354

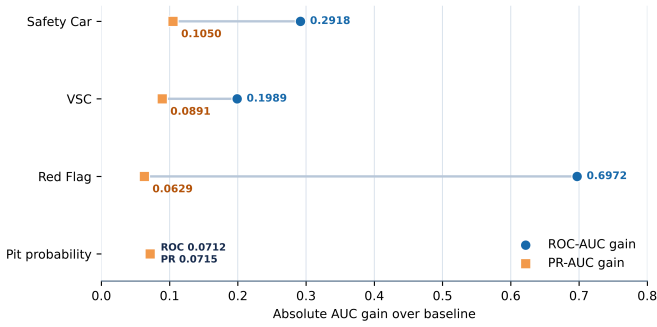


Fig. 5. Discrimination Gains Over Historical and Tyre-Age Priors

improved most, while the two traffic-related estimates also improved clearly. Exact values are provided on the bars and in Table IV.

The error reductions in Fig. 4 concern regression tasks. Disruption and pit-probability models require a separate view because their quality is assessed through class discrimination, particularly on rare events.

The filled circles in Fig. 5 show gains in general event discrimination, and the open squares show gains for the rare positive events. The Red Flag model gained most on the first measure, while the smaller gains for rare events reflect their limited number in the data.

The predictive comparisons do not describe every component of the system. Table V therefore records measured outputs for models without a direct baseline experiment.

Table V records the outputs for components without a direct baseline experiment. The compound-choice model frequently placed the observed compound within its two most likely choices, while the clean-air model showed useful discrimination and probability quality. Models 5A, 5B and 6 are simulation or recommendation components, so their values describe scale and output rather than predictive superiority. The reported Model 6 result is one example, not an aggregate claim.

A. Comparison to state of the art

The comparison with prior work is methodological rather than numerical. Dynamic programming research optimises pit timing under a specified race model [1]; Monte Carlo and digital-twin studies represent uncertainty through simulation [2], [3]; and Formula E and Formula 1 reinforcement-learning work learns a policy inside a simulator [4]–[8]. Those studies use different championships, inputs, target variables and evaluation settings. Their reported scores cannot therefore be compared directly with the MAE, ROC-AUC or PR-AUC values in Table IV.

This project differs in the sequence of evidence it provides. Pace, disruption, pit and compound behaviour, and traffic or rejoin cost are estimated separately from public driver-lap data before candidate actions are simulated. The ablation results then check whether those inputs mattered to the pipeline. Recent pace, circuit context, tyre state and local gaps each caused measurable deterioration when removed; Models 5 and 6 also changed selected actions when key simulation or risk inputs were removed. The supported claim is not universal superiority. It is that the project joins predictive, simulation and risk-aware decision stages in one inspectable workflow.

B. Ablation Study

Table VI summarises the controlled ablation results. The table reports a change in the appropriate metric after a feature group or component was perturbed; it is not a second baseline comparison. The evidence therefore explains which parts of the implemented pipeline contributed to a prediction or recommendation.

For Models 1–4, the ablation outcomes were consistent with the race mechanisms that the models were designed to represent. Recent pace dominated lap-time prediction. Circuit context was the strongest Safety Car contributor, whereas VSC ablations were less stable because VSC events were rarer and the logistic reference model reported convergence warnings. Pit timing depended most on tyre state, with race progress and pace or degradation next. Across clean-air, position-delta and rejoin-penalty tasks, track position and gaps were the

TABLE V
FINAL OUTPUTS OF MODELS WITHOUT A DIRECT BASELINE COMPARISON

Model	Final component	Measured output
M3B	XGBoost compound-choice model	Log loss: 0.7660; top-1 accuracy: 0.6493; top-2 accuracy: 0.9336.
M3C	Wet/intermediate compound handler	33,725 wet-handler rows; mean intermediate probability: 0.3802; mean wet-tyre probability: 0.2472.
M4A	XGBoost clean-air classifier	ROC-AUC: 0.7263; PR-AUC: 0.8862; Brier score: 0.1632.
M5A	Local strategy cost simulator	4,242 rows; 3,000 simulations; pitting immediately was best in 45.71% of evaluated rows.
M5B	Scenario-based Monte Carlo strategy simulator	5,000 simulations per run; outputs: mean finish, mean points, p_{top10} , p_{top5} , p_{podium} and p_{win} .
M6	Risk-aware recommendation layer	Example output: utility 6.2852; expected finish P6.41; expected points 7.17; confidence: Medium.

dominant group. These results support the use of pace, tyre state, circuit, and local-traffic features as distinct inputs rather than interchangeable generic race variables.

The component-level results also tested the integrated decision path. In Model 5A, removing pit-pressure or neutralisation signals changed more than half of the recommendations. In the selected Model 5B case, removing pace, pit and compound probabilities, pit-cost information, or wet/dry realism changed the recommended action. These results show sensitivity within the evaluated case; they do not prove that one strategy is best in every race.

Model 6 added a decision check rather than duplicating Model 5B. In the selected test case, the raw simulator slightly favoured waiting three laps and then pitting for medium tyres. Model 6 recommended remaining on the current tyres because the points difference was very small and this action had stronger compound confidence and lower downside risk. Removing confidence or risk-protection terms changed the selected action. This showed that Model 6 applied the stated risk-aware policy rather than simply repeating the raw simulation rank.

VII. DISCUSSION

The results show that better information was passed into the strategy simulator, especially for lap time, position change and rejoin cost. The 81.64% reduction in lap-time MAE and the 60.16% reduction in rejoin-penalty MAE were particularly useful. A pit stop is not credible as a recommendation if only its pace benefit is estimated; the traffic cost matters too.

The disruption and pit-probability models also improved on their prior-based baselines. Their ROC-AUC and PR-AUC results show that they supplied more useful probability estimates than historical rates alone. However, the Red Flag PR-AUC remained small because positive events were rare. The system consequently used disruption outputs to generate and weight scenarios, rather than treating a disruption as certain.

Models 5A, 5B and 6 connected these component predictions to a decision. Model 5A evaluated local costs across 4,242 rows, Model 5B produced 5,000 simulations per run, and Model 6 returned a ranked recommendation with expected finish, expected points and confidence. The main difference is that the system does not stop after predicting pace or pit

TABLE VI
SUMMARY OF CONTROLLED ABLATION RESULTS

Model	Key evidence	Meaning
M1	Recent pace: MAE +6.8872 s.	Current car rhythm was the strongest pace signal.
M2	Circuit context: Safety Car PR-AUC -0.1148.	Track-specific risk was central to interruption prediction.
M3	Tyre state: mean pit-timing PR-AUC -0.1184.	Tyre age, compound and stint state drove pit windows.
M3B-C	Race progress: dry-compound log loss +0.2047; no dry compound was the top choice on wet-handler rows.	Race distance guided dry choice; wet logic prevented implausible dry recommendations.
M4	Position and gaps: clean-air PR-AUC -0.1140; rejoin MAE +1.1680 s.	Local traffic and gap structure drove rejoin outcomes.
M5	Removing pit-pressure or neutralisation input changed 55.54% or 53.35% of recommendations.	Simulation depended on pit-window and interruption-risk inputs.
M6	Removing confidence or risk terms changed the selected test-case action.	The final layer added risk protection beyond the raw simulator rank.

behaviour. Instead, it combines those predictions with simulation before recommending an action. It compares feasible actions under the same race state, then reports both the raw simulated result and the final risk-aware recommendation. This demonstrates usable decision outputs; it does not prove that the chosen action is always better than the historical team decision or every competing action.

The ablation results strengthen this interpretation. They showed that the predictive models used behaviourally relevant signals: recent pace for lap time, circuit context for Safety Car risk, tyre state for pit timing, and position and gaps for traffic and rejoin effects. They also showed that the simulator and recommendation layer were sensitive to the upstream components they consumed. In particular, the change in Model 5 recommendations after removing pit-pressure or neutralisation inputs, and the Model 6 switch after removing confidence or risk terms, show that the final recommendation was not produced from a single opaque score.

LIMITATIONS

Public data do not contain the tyre temperatures, fuel corrections, power-unit modes, driver feedback or competitor radio information available to Formula 1 teams. Rare Red Flag events and changing wet-track conditions also limit model reliability. In addition, errors can accumulate when one model output is used by another part of the pipeline.

The reported baseline comparisons evaluate individual predictive components, while the ablation results test sensitivity within the evaluated data and selected simulation case. Neither evaluation establishes the true counterfactual optimum for every race state. VSC ablation results also require caution because the reference logistic model showed convergence warnings and VSC events were rare. The recommendation-change results for Models 5 and 6 are interpreted as evidence of component use and decision sensitivity, not universal proof that one strategy would outperform all alternatives.

VIII. CONCLUSION AND FUTURE WORK

This study developed a Formula 1 race strategy decision support system that combined Models 1–4, local pit-cost estimation, Monte Carlo simulation and a risk-aware recommendation layer. The predictive models improved on their selected baselines. Lap-time error fell by 81.64%, and rejoin-penalty error fell by 60.16%. The ablation study showed why these estimates mattered: recent pace, circuit context, tyre state and local traffic were not decorative inputs. When key simulation or risk-protection components were removed, the selected strategy changed.

The research contribution is how these estimates were used. Rather than treating pace, pit probability or disruption probability as separate final answers, the system merged them into one driver-lap state, estimated local pit and wait cost, simulated feasible actions and applied a role- and risk-aware recommendation. This addressed the integration gap in the literature by connecting optimisation, prediction and simulation in one traceable public-data decision-support workflow. The project is not claimed to be numerically superior to every cited study because their data, targets and evaluation settings differ. The supported conclusion is narrower: compared with the project baselines, the final models supplied better strategy inputs and the integrated system converted them into transparent, inspectable recommendations.

Future work can improve the system by incorporating richer telemetry, particularly tyre temperature, fuel correction, track evolution and real-time weather information. Circuit-specific modelling would also improve pit-loss, overtaking and tyre-behaviour estimates.

GENAI USAGE DISCLOSURE

Generative AI was used during the preparation of this dissertation as an editorial aid to improve grammar, clarity, sentence structure, and $\LaTeX$ formatting. Its use was limited to language refinement in the Abstract, Introduction, Conclusion, and general proofreading across the manuscript. It was not used to develop the research, methodology, experiments,

analysis, results, or conclusions. All AI-assisted edits were reviewed, verified, and accepted by the authors, who take full responsibility for the final content.

REFERENCES

- [1] O. F. Carrasco Heine and C. Thraves, “On the optimization of pit stop strategies via dynamic programming,” *Central European Journal of Operations Research*, vol. 31, no. 1, pp. 239–268, 2023, doi: 10.1007/s10100-022-00806-4.
- [2] A. Heilmeyer, M. Graf, J. Betz, and M. Lienkamp, “Application of Monte Carlo methods to consider probabilistic effects in a race simulation for circuit motorsport,” *Applied Sciences*, vol. 10, no. 12, Art. no. 4229, 2020, doi: 10.3390/app10124229.
- [3] A. Greasley, G. Panchal, and A. Samvedi, “The use of simulation with machine learning and optimization for a digital twin: A case on Formula 1 DSS,” in *Proc. Winter Simulation Conf.*, 2022, pp. 2198–2209, doi: 10.1109/WSC57314.2022.10015299.
- [4] X. Liu and A. Fotouhi, “Formula-E race strategy development using artificial neural networks and Monte Carlo tree search,” *Neural Computing and Applications*, vol. 32, no. 18, pp. 15191–15207, 2020, doi: 10.1007/s00521-020-04871-1.
- [5] X. Liu, A. Fotouhi, and D. J. Auger, “Formula-E race strategy development using distributed policy gradient reinforcement learning,” *Knowledge-Based Systems*, vol. 216, Art. no. 106781, 2021, doi: 10.1016/j.knsys.2021.106781.
- [6] X. Liu, A. Fotouhi, and D. J. Auger, “Formula-E multi-car race strategy development: A novel approach using reinforcement learning,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 9524–9534, 2024, doi: 10.1109/TITS.2024.3389155.
- [7] D. Thomas, J. Jiang, A. Kori, A. Russo, S. Winkler, S. Sale, J. McMillan, F. Belardinelli, and A. Rago, “Explainable reinforcement learning for Formula One race strategy,” in *Proc. 40th ACM/SIGAPP Symposium on Applied Computing*, 2025, pp. 1090–1097, doi: 10.1145/3672608.3707766.
- [8] D. Thomas, J. Jiang, A. Kori, A. Russo, S. Winkler, S. Sale, J. McMillan, F. Belardinelli, and A. Rago, “Race strategy reinforcement learning: Optimising pitstop strategy with emergent tactics in Formula One,” *Machine Learning*, vol. 115, Art. no. 146, 2026, doi: 10.1007/s10994-026-07081-3.
- [9] A. Heilmeyer, A. Thomaser, M. Graf, and J. Betz, “Virtual Strategy Engineer: Using artificial neural networks for making race strategy decisions in circuit motorsport,” *Applied Sciences*, vol. 10, no. 21, Art. no. 7805, 2020, doi: 10.3390/app10217805.
- [10] L. Paparusso, M. Riani, F. Ruggeri, and F. Braghin, “Competitors-aware stochastic lap strategy optimisation for race hybrid vehicles,” *IEEE Transactions on Vehicular Technology*, vol. 72, no. 3, pp. 3074–3089, 2023, doi: 10.1109/TVT.2022.3215171.
- [11] P. Dühr, D. Buccheri, C. Balerna, A. Cerofolini, and C. H. Onder, “Minimum-race-time energy allocation strategies for the hybrid-electric Formula 1 power unit,” *IEEE Transactions on Vehicular Technology*, vol. 72, no. 6, pp. 7035–7050, 2023, doi: 10.1109/TVT.2023.3237388.
- [12] S. Sasikumar, A. A. Leema, and P. Balakrishnan, “Data-driven pit stop decision support for Formula 1 using deep learning models,” *Frontiers in Artificial Intelligence*, vol. 8, Art. no. 1673148, 2025, doi: 10.3389/frai.2025.1673148.
- [13] F. Aguad and C. Thraves, “Optimizing pit stop strategies in Formula 1 with dynamic programming and game theory,” *European Journal of Operational Research*, vol. 319, no. 3, pp. 908–919, 2024, doi: 10.1016/j.ejor.2024.07.011.
- [14] J. F. Bekker and W. Lotz, “Planning Formula One race strategies using discrete-event simulation,” *Journal of the Operational Research Society*, vol. 60, no. 7, pp. 952–961, 2009, doi: 10.1057/palgrave.jors.2602626.
- [15] D. Piccinotti, A. Likmeta, N. Brunello, and M. Restelli, “Online planning for F1 race strategy identification,” in *Proc. Planning and Reinforcement Learning Workshop at ICAPS*, 2021.
- [16] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [17] T. Saito and M. Rehmsmeier, “The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets,” *PLoS One*, vol. 10, no. 3, Art. no. e0118432, 2015, doi: 10.1371/journal.pone.0118432.